

10.14 THE RENORMALIZATION SCALE AND THE RENORMALIZATION GROUP

Two complementary views of the same scale dependence: first the practical one, how the renormalization scale μ is chosen and varied in a calculation; then the conceptual one, in what sense the “renormalization group” is a group.

THE RENORMALIZATION SCALE μ IN THE $\overline{\text{MS}}$ SCHEME

In quantum field theories such as Quantum Chromodynamics (QCD), calculations often encounter infinities from quantum loop corrections in Feynman diagrams. To obtain finite, physical predictions, we employ renormalization, which absorbs these divergences into redefinitions of parameters like masses and coupling constants (e.g., the strong coupling α_s).

The $\overline{\text{MS}}$ (Modified Minimal Subtraction) scheme is a widely used renormalization method (§10.13). It subtracts only the divergent parts (poles in dimensional regularization) and certain constants. In this scheme, we introduce the renormalization scale μ , which at first appears unphysical since physical observables should not depend on our calculational choices. However, μ plays a vital role in capturing how parameters evolve with energy scales.

Consider μ as a “ruler” for measuring interactions at different energy “zoom levels.” In QCD, the strong coupling α_s is not constant but “runs” due to quantum effects, exhibiting asymptotic freedom: it is strong at low energies and weak at high energies. The renormalization group equation (RGE) describes this evolution:

$$\frac{d\alpha_s}{d\ln(\mu^2)} = -\beta(\alpha_s),$$

where, in this convention, $\beta(\alpha_s) = \beta_0\alpha_s^2 + \dots$ with $\beta_0 = (33 - 2n_f)/12\pi > 0$ for n_f quark flavors (note the opposite overall sign from $\beta = \mu dg/d\mu$ in §10.13). At one loop the solution is

$$\alpha_s(\mu) \approx \frac{\alpha_s(\mu_0)}{1 + \beta_0\alpha_s(\mu_0)\ln(\mu^2/\mu_0^2)}.$$

Here, α_s explicitly depends on μ . Measuring α_s at one scale (e.g., $\mu = M_Z \approx 91 \text{ GeV}$) allows prediction at others via the RGE. Measurements

of α_s at many different scales follow this running; Section 9.4 of the Particle Data Group's QCD review collects them in one graph.⁴ While μ is a technical parameter, it is dynamical, reflecting the physical scale of the process. In perturbative calculations, terms like $\ln(q^2/\mu^2)$ appear, where q is a momentum. In idealized processes with a single hard scale, such as the center-of-mass energy $\sqrt{s}$ in $e^+e^- \rightarrow \text{hadrons}$, we set $\mu^2 = s$ for several reasons:

1. **Minimizing Logarithms:** This makes $\ln(s/\mu^2) \approx 0$, reducing higher-order corrections, and so the error from truncating the series at finite order.
2. **Resumming Effects:** It incorporates some higher-order contributions into the running coupling.
3. **Physical Matching:** μ separates short- and long-distance physics, and the dominant scale of the process is where that separation belongs.

Choosing μ inappropriately can produce large logarithms, and, if μ is too small, a large α_s , breaking perturbation theory.

HOWEVER, $\mu^2 = s$ IS MERELY A “BEST-GUESS” CHOICE

Real physics calculations often involve multiple dynamical scales, such as relative momenta, particle masses, or kinematic variables. In these cases, setting $\mu^2 = s$ may not be optimal, as it could introduce large logarithms from other scales. Instead, we use physics intuition to select a “best-guessed” central scale μ_0 :

- Identify key scales (e.g., momentum transfer Q , jet p_T , masses m_q).
- Common choices: Geometric means like $\sqrt{Q_1 Q_2}$ for two scales, or dynamical options like $H_T/2$ (half the scalar sum of transverse momenta) in multi-jet events.
- In effective theories like SCET, multiple μ 's handle different sectors (hard, collinear, soft).

To validate, compute at higher orders (e.g., NLO or NNLO), where μ dependence should decrease as more terms cancel. Additionally, estimate uncertainties via scale variation: Vary μ around μ_0 (typically by factors of 1/2 to 2), and independently vary related scales like the factorization scale μ_F . The resulting band provides a conservative theoretical error estimate. Stability under variation indicates reliability; otherwise, higher orders or resummation may be needed.

This approach is iterative: Start with a reasonable μ_0 , compute, vary scales, compare orders, and refine based on the physics.

⁴J. Huston, K. Rabbertz and G. Zanderighi, “Quantum Chromodynamics,” in S. Navas *et al.* (Particle Data Group), Phys. Rev. D **110**, 030001 (2024), Sec. 9.4.

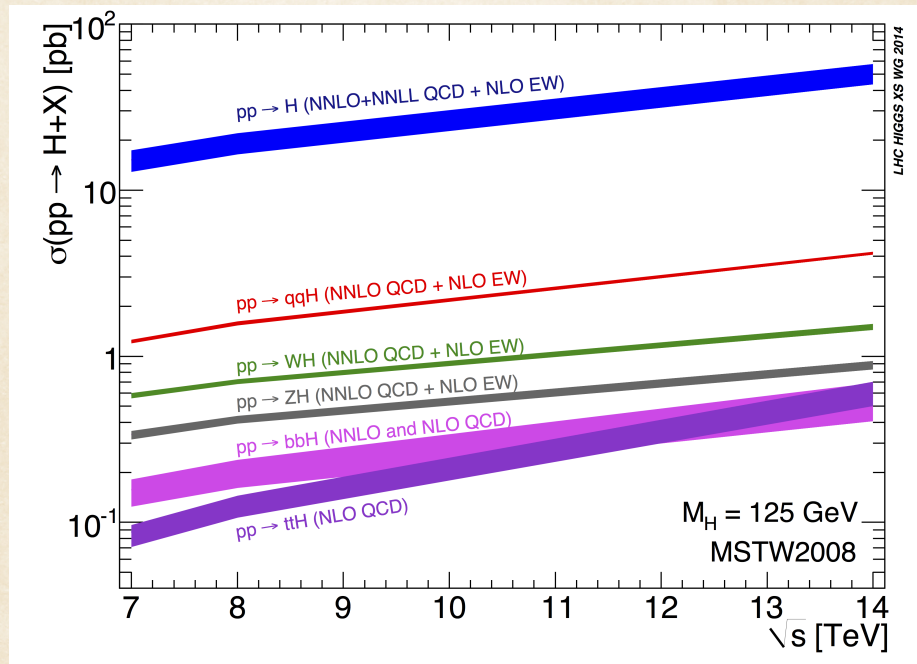


Figure 10.4: Higgs cross section as a function of center of mass energy of the collision. Part of the “band” is from uncertainty estimation via varying the renormalization (and factorization) scales.

IS THE “RENORMALIZATION GROUP” A GROUP?

One might wonder if the “Renormalization Group” (RG) is, in fact, a mathematical group. The name is a misnomer, and understanding why it’s both wrong and right reveals the entire conceptual leap from historical QFT to the modern, Wilsonian framework.

THE CENTRAL QUESTION: A GROUP OR NOT?

In the modern, arguably more general sense, the Renormalization Group is **not a mathematical group**, but a **semigroup**.

The confusion arises because the name is a historical artifact from an older, more formal perspective. To see the difference, we must distinguish between two different ways of thinking about the RG:

1. The modern (Wilsonian) picture of a *physical flow*.
2. The historical (QFT) picture of a *re-parameterization*.

THE MODERN ANSWER: A SEMIGROUP OF PHYSICAL FLOWS

When we think of the RG in the modern sense (thanks to Ken Wilson), we are describing a physical process of “coarse-graining” or “integrating out” high-energy (short-distance) degrees of freedom. We are literally averaging over and discarding information.

A mathematical **group** must satisfy four axioms:

- 2152 1. **Closure:** (Transform A) + (Transform B) = (New Transform C). (*This*
2153 *holds.*)
- 2154 2. **Associativity:** (A+B)+C = A+(B+C). (*This holds.*)
- 2155 3. **Identity Element:** A transformation that does nothing. (*This*
2156 *holds.*)
- 2157 4. **Inverse Element:** For every transformation, there must be an
2158 inverse that perfectly undoes it. (***This fails.***)

2159 The act of integrating out high-energy modes is an **irreversible** process.
2160 Once you've coarse-grained a system, you have *lost information* about the
2161 precise, short-distance microscopic arrangement.
2162 You cannot run the RG "backwards" from a low-energy effective theory to
2163 uniquely discover the one, true high-energy UV theory. In fact, the entire
2164 concept of **universality** rests on this fact: many different UV theories can
2165 "flow" to the *same* low-energy IR theory.
2166 This is a **many-to-one mapping**. Because the transformations are not
2167 invertible, the RG is not a group. It is a semigroup (or more specifically, a
2168 monoid, as it has an identity element). This irreversibility is what gives
2169 the RG "flow" its "arrow of scale."

2170 A DIFFERENT ASPECT: A LIE GROUP OF 2171 RE-PARAMETERIZATION

2172 So, why the name? The name is a historical artifact from the 1950s:
2173 Ernst Stueckelberg and André Petermann (1953) called it the "groupe de
2174 normalisation", and Gell-Mann and Low (1954) developed the same idea
2175 independently for QED at short distances.⁵
2176 Their "group" was not about integrating out physical modes. It was about
2177 the **ambiguity** in the perturbative renormalization procedure. (This is
2178 how we introduced renormalization in the last part of Chapter II of our
2179 classes.)

- 2180 • In perturbative QFT, to get rid of infinities, we introduce an
2181 arbitrary, unphysical energy scale, μ , often called the
2182 "renormalization scale" or "subtraction point."
- 2183 • All our renormalized couplings (g), masses (m), and field
2184 normalizations (Z) become functions of this arbitrary scale: $g(\mu)$,
2185 $m(\mu)$, $Z(\mu)$.
- 2186 • Physical observables (like a scattering cross-section) *cannot* depend
2187 on this arbitrary choice of μ .
- 2188 • Therefore, a change in our choice of scale (from μ_1 to μ_2) must be
2189 perfectly compensated by a corresponding change in the couplings
2190 and fields to keep the physics invariant.

⁵E. C. G. Stueckelberg and A. Petermann, *Helv. Phys. Acta* **26**, 499 (1953); M. Gell-Mann and F. E. Low, "Quantum electrodynamics at small distances," *Phys. Rev.* **95**, 1300 (1954).

2191 This set of transformations is a group.

- 2192 • **Closure:** A transformation $\mu_1 \rightarrow \mu_2$ followed by $\mu_2 \rightarrow \mu_3$ is just the
2193 transformation $\mu_1 \rightarrow \mu_3$.
- 2194 • **Associativity:** This holds.
- 2195 • **Identity:** The transformation $\mu_1 \rightarrow \mu_1$.
- 2196 • **Inverse:** The transformation from $\mu_2 \rightarrow \mu_1$ is the *exact* inverse of
2197 $\mu_1 \rightarrow \mu_2$.

2198 This forms a continuous, invertible **Lie group**: a one-parameter group,
2199 the additive group of $\ln \mu$. The Callan–Symanzik and Gell-Mann–Low
2200 equations are the differential expressions (i.e., the Lie algebra) of this
2201 group. They describe how the couplings must “run” with scale μ to
2202 preserve the invariance of the physics.

2203 RECONCILIATION: IT IS UP TO THE PHYSICS CONTEXT

2204 One can rightly ask: “Even if I am given a single UV theory, like the
2205 Standard Model, its flow is described by a deterministic differential
2206 equation (the Callan-Symanzik equation). Can’t I just run this equation
2207 backwards and find the UV theory? Isn’t that an inverse?”

2208 That’s correct. If you are given **one single, well-defined theory**, the flow
2209 is just a **re-parameterization** of that one theory. You can slide your
2210 “viewing window” μ up and down, and the transformations are invertible.
2211 In this limited sense, for a *single theory*, the flow is a group.

2212 The Wilsonian revolution was a philosophical shift. We stopped thinking
2213 about a *single theory* (in particular, for condensed matter theorists) and
2214 started thinking about the **space of all possible theories** (so-called
2215 “theory space”).

2216 The physical act of coarse-graining is a *flow* in this giant space. Each
2217 point in the space is a different Lagrangian (a different set of all possible
2218 coupling constants).

2219 In this picture, we see universality. For example:

- 2220 • An Ising model on a simple cubic lattice.
- 2221 • An Ising model on a body-centered cubic lattice.
- 2222 • A real-world liquid–gas system near its critical point.

2223 These are three *wildly different* microscopic (UV) theories. But as we
2224 apply the Wilsonian RG (zoom out, coarse-grain, and rescale), they all
2225 flow toward the *exact same* point in theory space: the three-dimensional
2226 “Ising fixed point.”

2227 This **many-to-one** map has no inverse, which is why the RG is only a
2228 semigroup. We prefer this picture because it *explains the physics*: *why*
2229 our low-energy world looks simple.

2230 The irreversible, information-losing, semigroup nature of the RG flow is
2231 what washes away all the complex, unknown details of Planck-scale
2232 physics, leaving us with the simple, renormalizable, effective field
2233 theories (like the Standard Model) that we use to describe our world.